Chapter 5 – Roots of Equations

To solve $f(x) = ax^2 + bx + c = 0$, $x = \underbrace{\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$ quadratic formula

- Roots are also called as zeros of the function.
- There are many functions for which roots can't be found easily. $\Rightarrow$ use numerical methods.
- e.g: $f(x) = e^{-x} - x = 0$ can't be solved analytically.
- e.g: Falling parachutist

$$v(t) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right) \qquad (1)$$

(c: drag coeff, m: mass)

goal: determine the drag coefficient, given m, to attain a prescribed velocity in a set time period.

Eq^n (1) can't be solved analytically. There's no way to isolate c on one side of the eq^n.

use a numerical method: Reexpress (1)

$$f(c) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right) - v$$, g, m, t, v are given (known)

Solve $f(c) = 0$ $\Rightarrow$ c is the root of the eq^n.

Mathematical Background

Algebraic Equations: involve algebraic operations (addition, subtraction, multip., division, power, square root etc)

$$f(x) = \frac{\sqrt{1+x^3}}{x^{3/7} - \sqrt{7}x^{1/3}},$$

- Polynomial are algebraic functions:

$$f_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \quad , n: \text{order of polynomial.}$$

(a_0, a_1, a_2: constants)

Transcendental Functions: non-algebraic. Include trigonometric, exponential, logarithmic etc. functions

$$f(x) = \ln x - 1 \quad , f(x) = e^{-0.2x} \sin(3x - 0.5)$$

Two main Problems

1) Determine the real roots of algebraic & transcedental functions.

2) Determine all real and complex roots of polynomials.

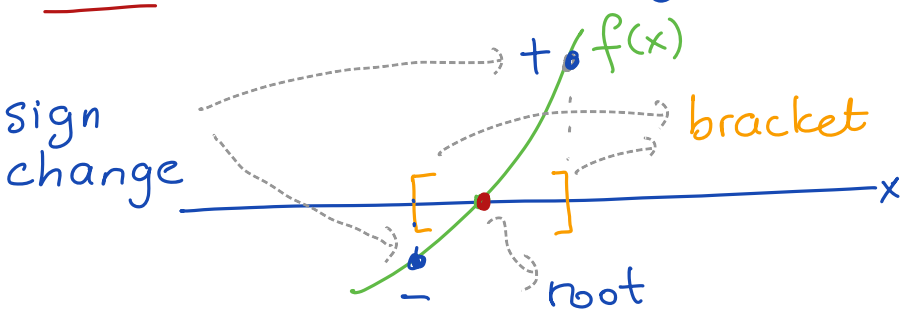
Preview of Finding Roots

1) Bracketing methods: (start with a range that contains the root) — two initial guesses
(Ch5)
- Bisection
- False position — always converge
- Graphical method — slow to "

2) Open methods (the initial guess don't bracket the root)
(Ch6)
- Computationally more efficient, do not always converge.
- one-pt iteration
- Newton-Raphson
- Secant Method

Bracketing Methods

Fact: a function changes sign in the vicinity of a root

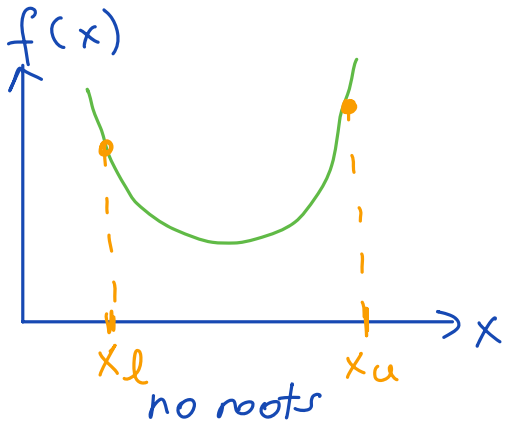


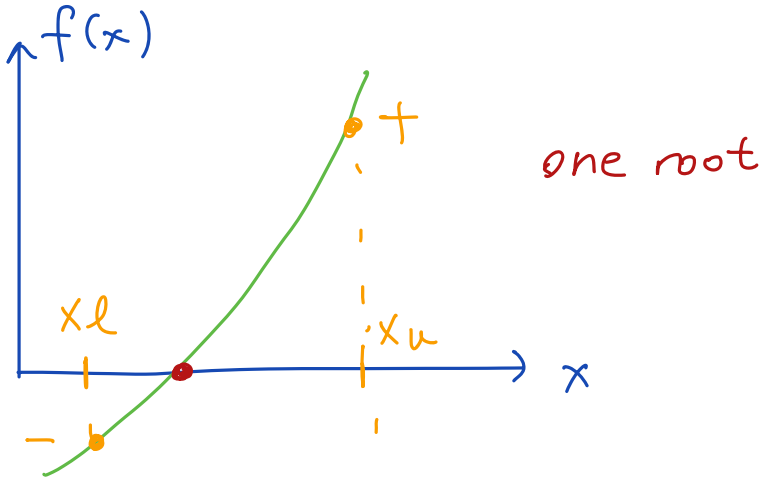
2 initial guesses that bracket the root.

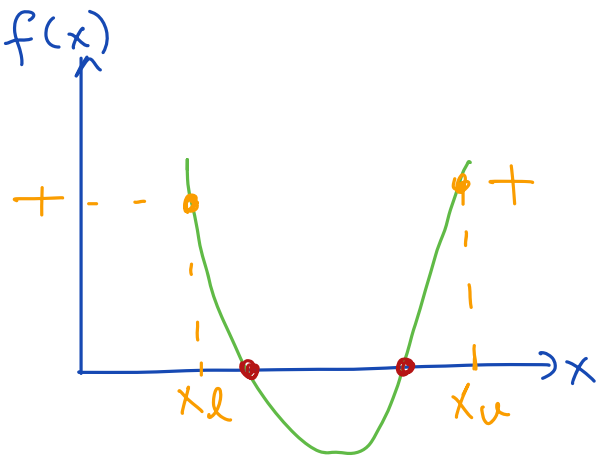
Graphical Method: To solve $f(x)=0$, plot $f(x)$ and observe where it crosses the x-axis.

→ Check the validity of the root by substituting the eqn: $f(x) \cong 0$

- are not precise

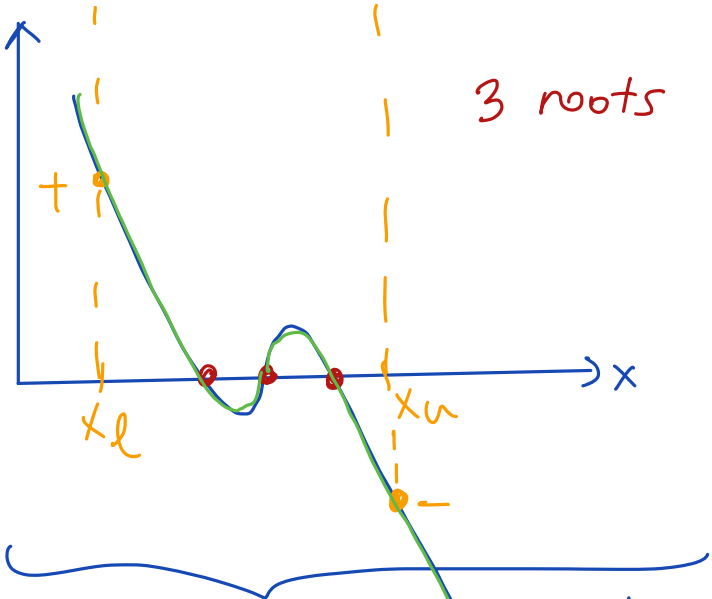






even # of roots between $[x_l, x_u]$

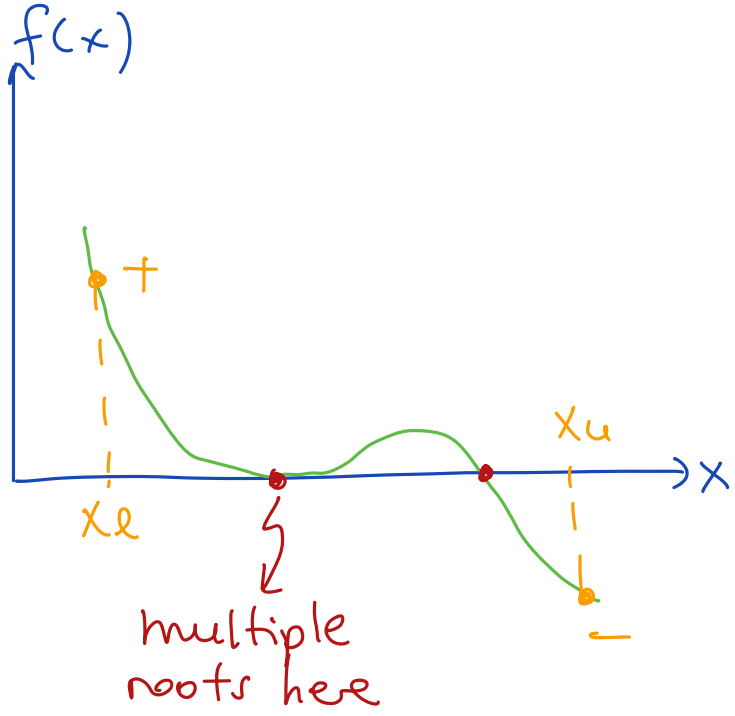
$f(x_l)$ and $f(x_u)$ have the same sign



$f(x_l)$ and $f(x_u)$ have opposite signs

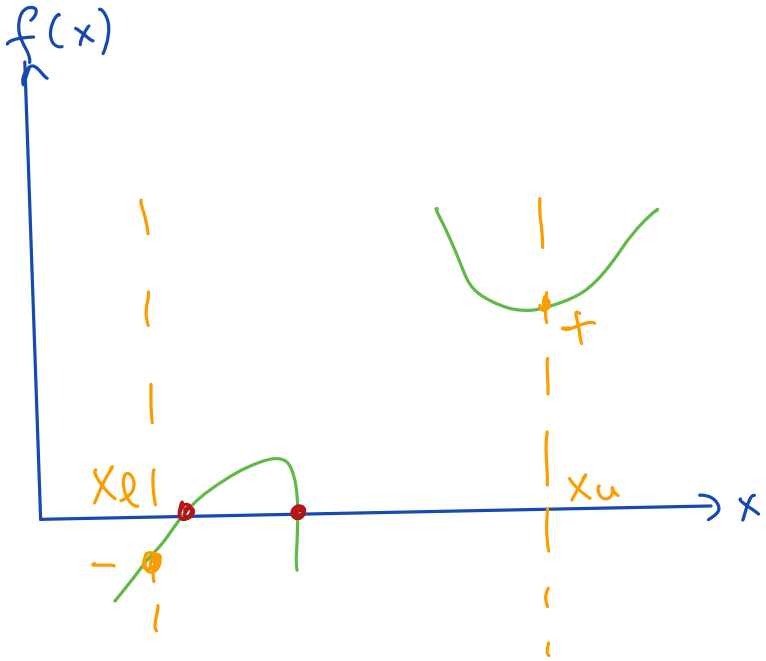
(odd number of roots in $[x_l, x_u]$)

Exceptions:



e.g: $(x-2)^2(x-4)$

multiple root at $x = 2$



Discontinuous Functions

even # of roots

The Bisection Method

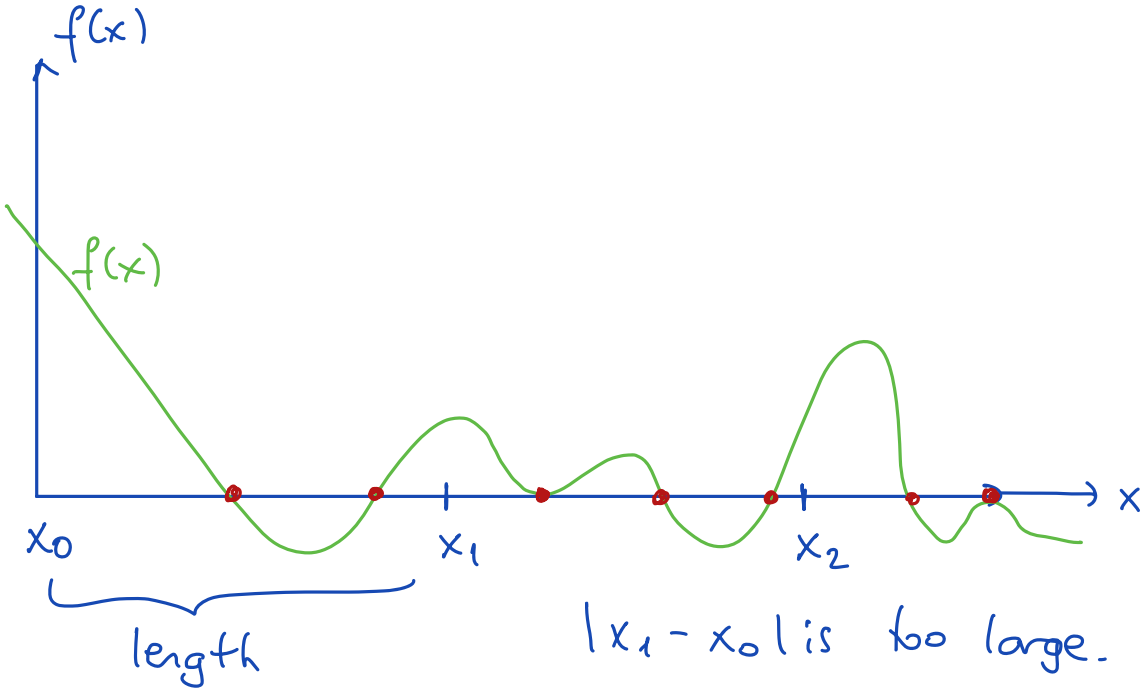
if $f(x_\ell)$ and $f(x_u)$ have opposite signs

$$f(x_\ell)\, f(x_u) < 0$$

then there is at least one real root between x_ℓ and x_u.

⇒ How do we find x_ℓ and x_u?

use incremental search:

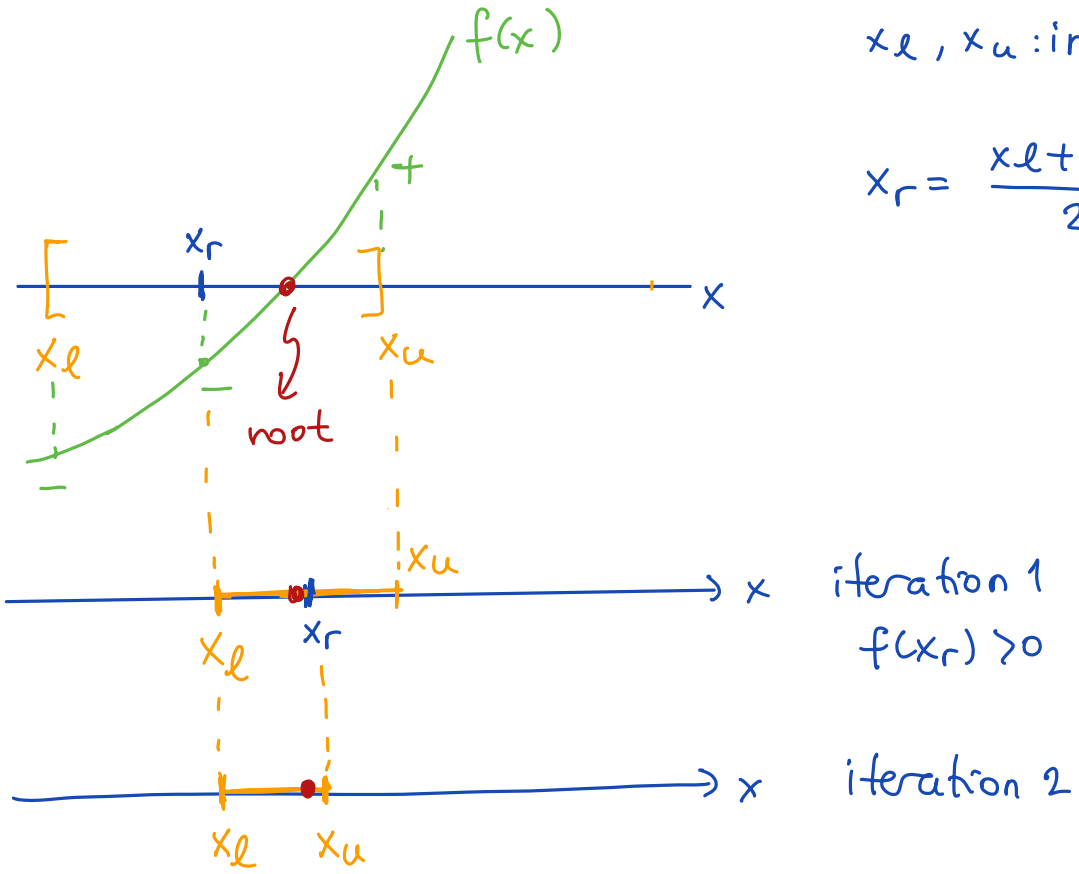


$|x_1 - x_0|$ is too large.

if length is small ⇒ search is time consuming

" is too large ⇒ closely spaced roots may be missed.

Bisection Method (Binary Chopping, interval halving)



x_l, x_u : initial interval

$x_r = \frac{x_l + x_u}{2}$: root estimate

Step 1 Choose x_l and x_u s.t. $f(x_l) f(x_u) < 0$

Step 2 Estimate the root as: $x_r = \frac{x_l + x_u}{2}$

Step 3 If $f(x_l) f(x_r) < 0 \Rightarrow$ root is in the lower interval

set $x_u = x_r$, return to step 2

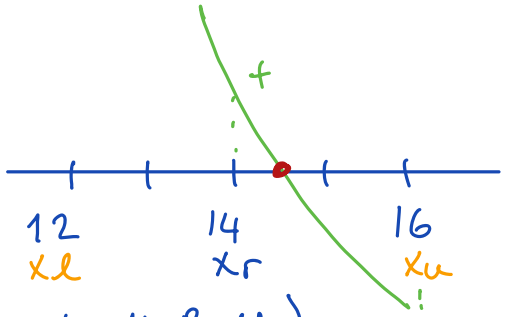
If $f(x_l) f(x_r) > 0 \Rightarrow$ root is in the upper interval

set $x_l = x_r$, return to step 2

If $f(x_l) f(x_r) = 0$, root $= x_r$, terminate

or If $\varepsilon_a < eps$ stop $\varepsilon_a = \left| \frac{x_r^{new} - x_r^{old}}{x_r^{new}} \right| 100\ \%$

Ex



(True root : 14.8011)

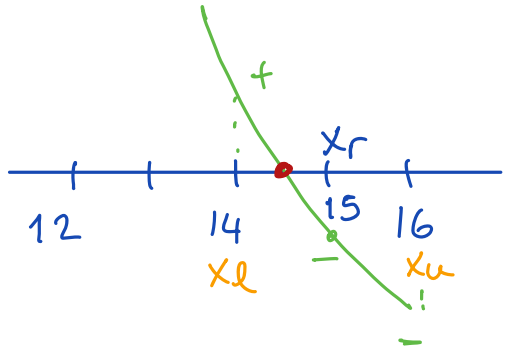
iteration 1 :

- $x_l = 12$, $x_u = 16$, $x_r = \frac{12+16}{2} = 14$
- $f(x_l) f(x_r) > 0 \Rightarrow x_l = 14 = x_r$
- $\varepsilon_t = 5.3\%$ (true percent relative error)

iteration 2 : $(x_r^{old} = 14)$

- $x_l = 14$, $x_u = 16$, $x_r^{new} = \frac{14+16}{2} = 15$
- $f(x_l) f(x_r) = f(14) f(15) < 0$

  $\Rightarrow x_u = x_r = 15$
- $\varepsilon_t = 1.3\%$



(3) $x_l = 14$, $x_u = 15$, $x_r = \frac{14+15}{2} = 14.5$ $\Rightarrow \varepsilon_t = 2\%$

$f(x_l) f(x_r) = f(14) f(14.5) > 0 \Rightarrow x_l = x_r$

repeat until ε_t is sufficiently small

OR " " convergence

After 6 iterations , let $\varepsilon_s = 0.5\%$ (predefined error value)

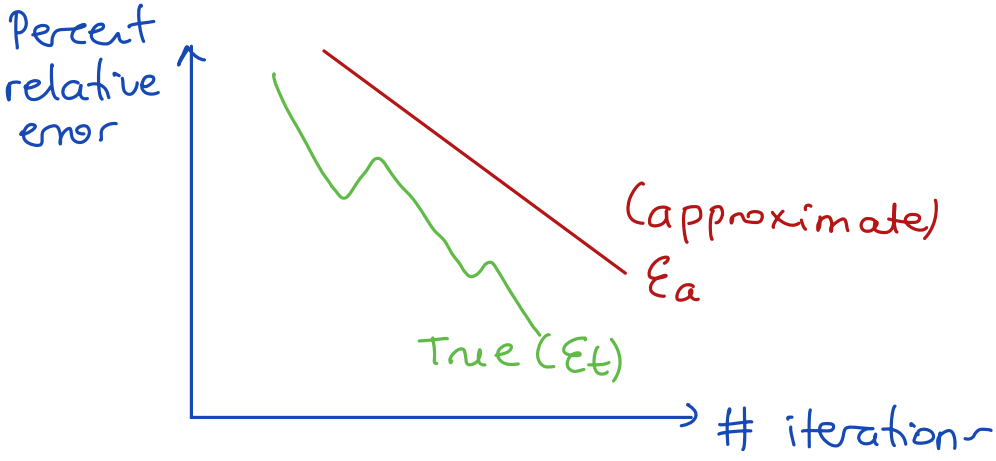
$x_l = 14.75$, $x_u = 14.875$, $x_r = 14.8125$

$$\varepsilon_a = \left| \frac{x_r^{new} - x_r^{old}}{x_r^{new}} \right| 100 = 0.422\%$$

Since $\varepsilon_a < \varepsilon_s$

$\Rightarrow$ stop iterations.

- Error (ε_t) does not decrease at each iteration

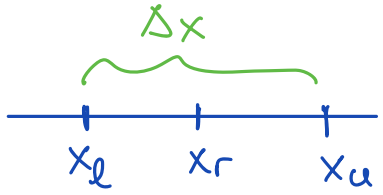


- Since $\varepsilon_a > \varepsilon_t$

  if $\varepsilon_a < \varepsilon_s \Rightarrow \varepsilon_t < \varepsilon_s$

- ε_a will always be greater than ε_t for bisection method:

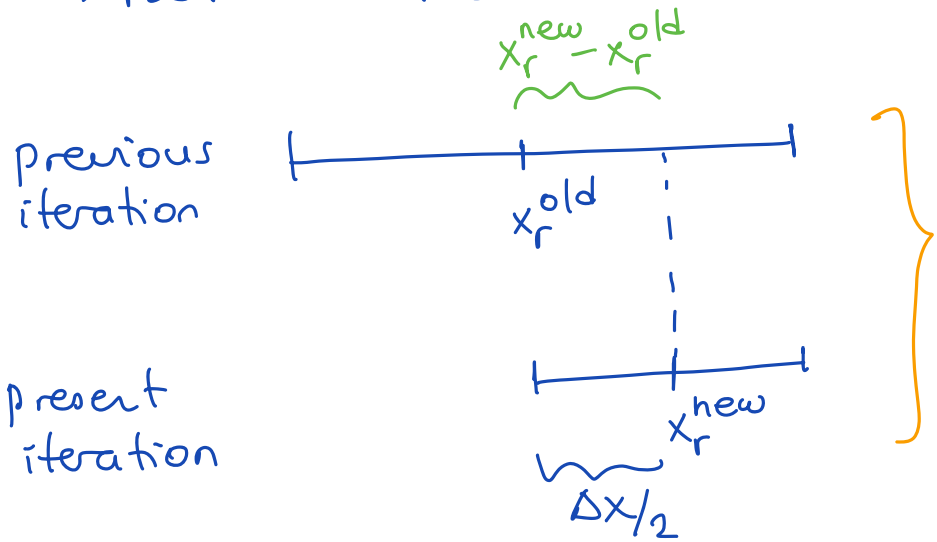
  if $x_r = \frac{x_l + x_u}{2}$ (root estimate)



the true root lies somewhere within an interval of

$$\frac{x_u - x_l}{2} = \frac{\Delta x}{2}$$

$\Rightarrow$ root is somewhere within $\pm \frac{\Delta x}{2}$ of estimate (x_r)

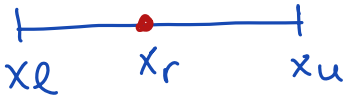


$x_r^{new} - x_r^{old} = \frac{\Delta x}{2}$ is an exact upper bound on true error.

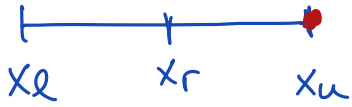
The root is always in the bracketing interval

Three ways in which an interval may bracket the root

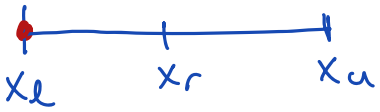
True value is at the center:

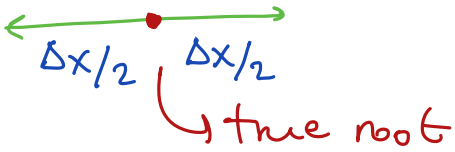


True value at x_u :



True value at x_l :





- Calculate # iterations required to achieve an absolute error: (before starting the iterations)

iteration 0 : $E_a^0 = x_u^0 - x_l^0 = \Delta x^0$

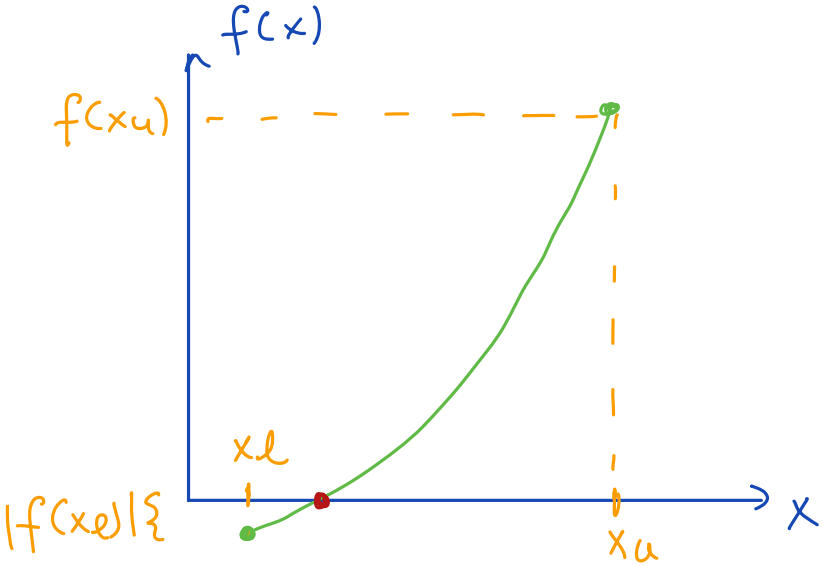
iteration 1 : $E_a^1 = \frac{\Delta x^0}{2}$

iteration n : $E_a^n = \frac{\Delta x^0}{2^n} < E_{a,d}$ (desired error)

$$n = \frac{\log(\Delta x^0 / E_{a,d})}{\log 2} = \log_2\left(\frac{\Delta x^0}{E_{a,d}}\right)$$

The False Position Method

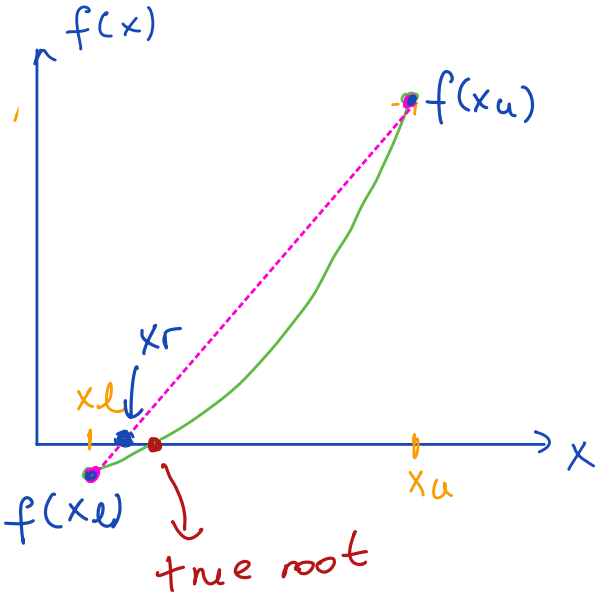
Shortcoming of Bisection: in diving the interval $[x_l, x_u]$ into half it doesn't take into account the magnitudes $f(x_l)$ and $f(x_u)$. If $f(x_l) < f(x_u)$ it is more likely that the root is closer to x_l than x_u.



root is closer to x_l

$$|f(x_l)| < f(x_u)$$

Solution: join $f(x_l)$ and $f(x_u)$ by a straight line. The interseection of this line with x-axis is an improved root estimate.



use similar triangles

$\triangle x_l x_r f(x_l) \sim \triangle x_r x_u f(x_u)$

$$\frac{f(x_l)}{x_r - x_l} = \frac{f(x_u)}{x_r - x_u}$$

derivation in the book

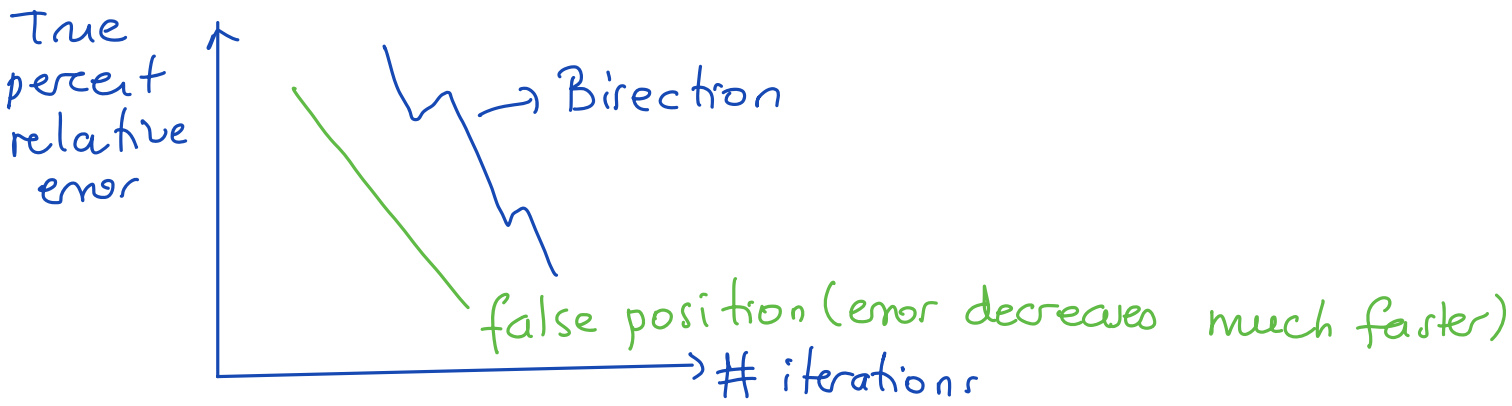
$$x_r = x_u - \frac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$$

False-position formula (Regula-falsi) (linear interpolation)

→ The value x_r replaces, x_l or x_u, whichever give a function value with the same sign as $f(x_r)$.

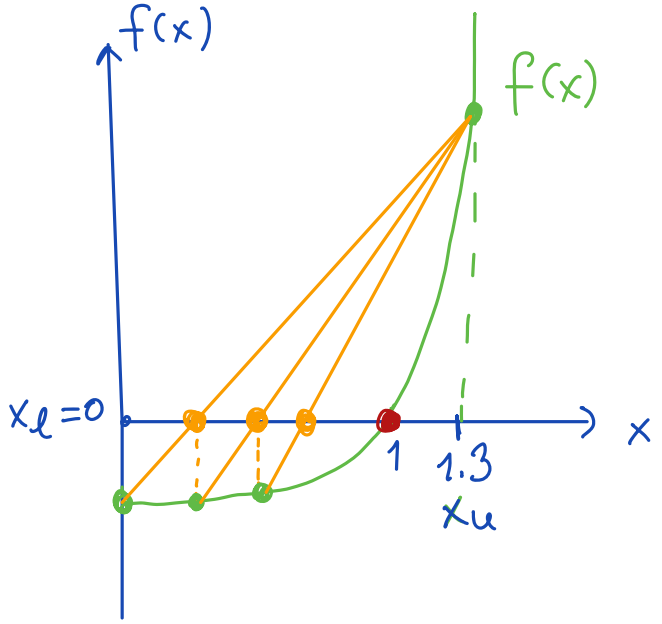
→ The values x_l & x_r always bracket the root

→ Repeat process until an ϵ_a (approximate relative error) is small.



EX A case when bisection is preferable to False Position

Locate the root of: $f(x) = x^{10} - 1$ between $x=0$ and $x=1.3$



- Slow convergence of false position
- The function violates the observation upon which false-position is built on: if $f(x_l)$ is much closer than to zero as compared to $f(x_u)$, the root is closer to x_l than x_u.

iteration = 5

bisection: $\epsilon_t = 1.6\%$, $\epsilon_a = 4.0\%$

false-position: $\epsilon_t = 59.2\%$, $\epsilon_a = 77.1\%$

- Check your result by substituting the root in $f(x)$ and check $f(x_r) \approx 0$